Reward Recovery in Sequential Decision Models

Pranjal Rawat*

April 2026

1 Setup

Consider a stationary infinite-horizon MDP with finite state space $\mathcal{S}$, finite action space $\mathcal{A}$, discount factor $\beta \in (0,1)$, transition kernel $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, and unknown reward $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$. Additive T1EV shocks yield the soft Bellman recursion.

$$Q^*(s,a) = r(s,a) + \beta \sum_{s'} P(s' \mid s,a)\, V^*(s'), \tag{1}$$

$$V^*(s) = \log \sum_{a'} \exp Q^*(s,a'), \tag{2}$$

$$\pi^*(a \mid s) = \exp\bigl(Q^*(s,a) - V^*(s)\bigr). \tag{3}$$

The advantage $A^*(s,a) = Q^*(s,a) - V^*(s) = \log \pi^*(a \mid s)$ is identified from CCPs. The reward r is not identified in levels: any $\tilde{r}(s,a) = r(s,a) + \lambda(s)$ generates identical CCPs (Magnac and Thesmar, 2002). The set of all such $\tilde{r}$ is the identified set (Kalouptsidi et al., 2021).

2 Five Approaches

2.1 Reduced-form Q

The reduced-form approach estimates $Q_\omega(s,a)$ by logit MLE.

$$\hat{\omega} = \arg\max_{\omega} \sum_{i,t} \log \frac{\exp Q_\omega(s_{it}, a_{it})}{\sum_{a'} \exp Q_\omega(s_{it}, a')}. \tag{4}$$

At the population optimum, $Q_{\hat{\omega}}(s,a) = Q^*(s,a) + c(s)$ for some action-independent c, so the backed-out "reward" is $A^*(s,a)$, not $r(s,a)$. No Bellman equation is imposed and no transition data are used.

2.2 AIRL

The AIRL estimator of Fu et al. (2018) uses the following discriminator decomposition.

$$f_\theta(s,a,s') = g_\theta(s,a) + \beta\, h_\phi(s') - h_\phi(s). \tag{5}$$

*Georgetown University. pp712@georgetown.edu.

Here g_θ approximates the reward and $\beta h(s') - h(s)$ is potential-based shaping (Ng et al., 1999). At the GAN optimum the discriminator score equals the advantage, $f^* = A^*$ (Fu et al., 2018, Appendix A.4), which implies the following expression for the recovered reward network.

$$g^*(s,a) = r(s,a) - \delta(s) + \beta \sum_{s'} P(s' \mid s,a)\, \delta(s'), \tag{6}$$

where $\delta(s) \equiv V^*(s) - h^*(s)$ is the shaping residual. Unless $\delta \equiv 0$, the recovered g^* is a potential-based perturbation of r that generates the same policy in the original MDP but can produce different behavior under changed dynamics.

Fu et al. address this through a state-only restriction: if g_θ depends only on s (not a) and the dynamics satisfy a decomposability condition, then $g^*(s) = r(s) + \text{const}$, recovering a disentangled reward that transfers across dynamics. This is what enables their sim-to-real experiments. However, the state-only restriction requires $r(s,a) = r(s)$, which fails whenever rewards depend on actions (e.g., paying vs. waiting).

2.2.1 LSW anchor identification

Lee et al. (2026) (henceforth LSW) take a different approach. They allow action-dependent rewards and impose two anchors instead. First, the exit payoff is normalized to zero, so $g(s, \text{exit}) = 0$ for all s. Second, the shaping function is pinned at the absorbing state, so $h(s_{\text{abs}}) = 0$, where exit deterministically transitions to s_{abs}.

Lemma 1 (Non-identifiability)**.** *If (g,h) and $(\bar{g},\bar{h})$ produce the same f for all (s,a,s'), then $\bar{h} = h + \Phi$ and $\bar{g}(s,a) = g(s,a) + \Phi(s) - \beta\Phi(s')$ for some $\Phi : \mathcal{S} \to \mathbb{R}$.*

Proof. Rearrange $g + \beta h(s') - h(s) = \bar{g} + \beta\bar{h}(s') - \bar{h}(s)$ and define $\Phi = \bar{h} - h$. □ □

Theorem 2 (Anchor identification)**.** *Under the two anchors, if (g,h) and $(\bar{g},\bar{h})$ realize the same f, then $g = \bar{g}$ and $h = \bar{h}$.*

Proof. By Lemma 1, $\bar{g}(s, \text{exit}) = 0 + \Phi(s) - \beta\Phi(s_{\text{abs}})$. The exit anchor forces $\Phi(s) = \beta\Phi(s_{\text{abs}})$ for all s. The absorbing anchor forces $\Phi(s_{\text{abs}}) = 0$. Hence $\Phi \equiv 0$. □ □

Theorem 3 (Recovery of true reward)**.** *If the true pair (r^*, V^*) satisfies $r^*(s, \textit{exit}) = 0$ and $V^*(s_{\text{abs}}) = 0$, then any AIRL pair (g,h) realizing A^* under the same anchors satisfies $g = r^*$ and $h = V^*$.*

Proof. Because (g,h) realizes A^*, for every realized transition (s,a,s') we have

$$g(s,a) + \beta h(s') - h(s) = r^*(s,a) + \beta V^*(s') - V^*(s).$$

Define $\Phi = h - V^*$. Then

$$g(s,a) - r^*(s,a) = \Phi(s) - \beta\Phi(s').$$

For the exit action, $s' = s_{\text{abs}}$ deterministically, and both the true reward and the recovered reward satisfy the exit anchor. Hence

$$0 = g(s, \text{exit}) - r^*(s, \text{exit}) = \Phi(s) - \beta\Phi(s_{\text{abs}}).$$

The absorbing-state anchor applied to both h and V^* gives $\Phi(s_{\text{abs}}) = 0$, so $\Phi(s) = 0$ for all s. Therefore $h = V^*$, and substituting back gives $g = r^*$. □ □

The ERM framework of Kang et al. (2025) uses the same anchor principle but solves a single non-adversarial optimization instead of a GAN.

The LSW model estimates segment-specific reward and policy functions for K latent consumer types. Each user belongs to segment $z \in \{1, \ldots, K\}$, and the objective is to recover the segment-specific reward $r^*(s, a, z)$ and policy $\pi^*(a \mid s, z)$ that rationalize observed trajectories under the entropy-regularized Bellman equation. Estimation proceeds via an EM algorithm: the E-step computes posterior segment membership from trajectory likelihoods, and the M-step updates each segment's reward via AIRL with the anchor constraints applied per segment.

2.3 IQ-Learn

The IQ-Learn estimator of Garg et al. (2021) uses the inverse Bellman operator $r_Q(s, a) \equiv Q(s, a) - \beta \sum_{s'} P(s' \mid s, a)\, V_Q(s')$, which expresses reward as a function of Q alone. IQ-Learn estimates a single Q-function by solving the following optimization problem.

$$\max_Q \; \mathbb{E}_{(s,a)\sim\rho^*}\big[\phi\big(r_Q(s, a)\big)\big] - (1 - \beta)\, \mathbb{E}_{s_0\sim\mu_0}\big[V_Q(s_0)\big], \tag{7}$$

The function $\phi(x) = x - x^2/2$ corresponds to χ^2 divergence. At the optimum, r_Q separates immediate reward from continuation values via the Bellman constraint. IQ-Learn requires the transition kernel $P(s' \mid s, a)$ to compute the inverse Bellman operator and uses it in the objective through $V_Q(s')$. It does not require adversarial training.

2.4 GLADIUS

The GLADIUS estimator of Kang et al. (2025) parameterizes both $Q_\theta(s, a)$ and $\hat{E}_\theta[V(s') \mid s, a]$ with neural networks, trained jointly on observed (s, a, s') transitions via the following loss.

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_i \log \pi_\theta(a_i \mid s_i) + \lambda \frac{1}{N} \sum_i \big(\hat{E}_\theta[V(s'_i) \mid s_i, a_i] - V_\theta(s'_i)\big)^2, \tag{8}$$

where $V_\theta(s') = \log \sum_{a'} \exp Q_\theta(s', a')$. After training, reward is extracted via the same inverse Bellman operator as IQ-Learn. Unlike IQ-Learn, GLADIUS does not require the transition matrix P at estimation time. It learns $\hat{E}[V(s') \mid s, a]$ from empirical next-state observations rather than computing it from known transitions. The MLP architecture shares parameters across states, providing implicit regularization that tabular Q parameterizations lack.

2.5 NFXP

The nested fixed point estimator of Rust (1987) parameterizes the flow utility $u_\theta(s, a)$, solves the Bellman equation exactly under known transitions $P(s' \mid s, a)$, and maximizes the log-likelihood of observed actions via the implied softmax probabilities. We use a policy/value iteration hybrid to solve the inner fixed point and a BHHH/BFGS outer loop. Under correct specification, NFXP consistently recovers r^* and yields valid Type B counterfactuals. Its trade-off is computational cost.

3 Counterfactual Taxonomy

Definition 4 (Type A: state extrapolation)**.** *The transition kernel P and reward r are unchanged. Only the realized state values shift. The counterfactual CCPs require only evaluating the estimated Q at new state inputs, with no Bellman re-solve.*

Definition 5 (Type B: environment change)**.** *The reward r is unchanged, but the transition kernel shifts to $\tilde{P}$ or the action set changes. The counterfactual Q must satisfy a new Bellman equation $\tilde{Q} = r + \beta\tilde{P}\tilde{V}$, which requires the structural reward r separated from the old continuation values.*

Proposition 6 (Type A equivalence)**.** *Under Type A, all methods considered yield identical counterfactual CCPs.*

Proof. Each method recovers scores whose within-state action differences match Q^*, which can be written as $Q^*(s, a) + c(s)$ for some action-independent c. For any shifted state $\tilde{s}$, the softmax $\pi(a \mid \tilde{s}) \propto \exp(Q^*(\tilde{s}, a) + c(\tilde{s}))$ cancels $c(\tilde{s})$. □ □

Proposition 7 (Type B divergence)**.** *Under Type B, reduced-form and AIRL-without-anchors produce invalid predictions. AIRL with anchors, IQ-Learn, and GLADIUS produce valid predictions when the structural reward is correctly recovered.*

The reduced-form estimate $Q^* + c(s)$ bundles r with βPV^*, so under $\tilde{P}$ it cannot disentangle r. AIRL without anchors recovers $g^* = r + \text{shaping}$, which is observationally equivalent in the original MDP but produces incorrect behavior under $\tilde{P}$ unless $\delta \equiv 0$. AIRL with anchors recovers $g^* = r^*$ exactly (Theorem 3). IQ-Learn and GLADIUS separate r via the inverse Bellman operator. The LSW anchors provide what Kalouptsidi et al. (2021) call the "additional restrictions" needed to identify counterfactuals under transition changes.

4 Application to LSW (2026)

LSW study sequential content consumption with three actions (buy, wait, exit) using AIRL under the two anchors and deterministic transitions. Their wait-time validation (Section 6.3) is a Type A counterfactual, so every method that recovers the advantage function would pass it identically. Their wait-for-free (WFF) removal and segment-specific delay policies (Sections 7.1–7.2) are Type B because they change the action-set structure and the MDP that each segment faces. Their content-based paywall (Section 7.3) is also Type B because high-content episodes become paid-only, altering the transition structure episode by episode.

5 Simulation Study

Data sources. Analyses A–C and E–G report population-level results computed from the solved MDP or from closed-form shaping, with no estimation noise. The JSON artifact for these analyses is `experiments/identification/results/full_simulation.json`, which also drives the repository figures. Analysis D reports finite-sample outcomes using the estimator implementations in `econirl`.

We construct a tabular MDP that mimics the LSW book-reading environment. The state encodes episode position, content quality, and wait-time level, giving $15 \times 3 \times 4 = 180$ regular states plus one absorbing state. The three actions are buy, wait, and exit, with deterministic transitions and $\beta = 0.95$. The true reward is $r^*(s, \text{buy}) = u(c) - 2.0$, $r^*(s, \text{wait}) = u(c) - 0.08w$, and $r^*(s, \text{exit}) = 0$, where $u \in \{0.5, 1.5, 3.0\}$. Analyses A through C and E through G are evaluated at the population level with no estimation noise. Analysis D uses finite-sample estimation with the actual IQ-Learn and GLADIUS implementations from the `econirl` package.

5.1 A. Reward recovery

Methods with anchors or Bellman consistency pin the value function offset and recover r^* exactly. The reduced-form estimator converges to A^*, which bundles reward with continuation values. AIRL without anchors recovers a shaped perturbation with $\delta = 0.5V^*$. All five non-NFXP methods produce Type A CCP errors below 10^{-6}, confirming Proposition 6.

Method	MSE vs. r^*	Correlation with r^*
Oracle / AIRL with anchors / IQ-Learn / GLADIUS	0.000	1.000
AIRL without anchors	10.005	0.440
Reduced-form	40.020	0.138

Table 1: Population-level reward recovery.

5.2 B. Type B error vs. transition shift

The counterfactual changes the transition kernel so that buying advances k episodes instead of one. The Bellman equation is re-solved under each method's recovered reward. The reduced-form approach cannot separate reward from continuation values, so it re-solves with A^* as a proxy for reward, which embeds the original transition structure. AIRL without anchors re-solves with the shaped reward g^*, which embeds the shaping residual.

k	Oracle	AIRL with anchors	IQ-Learn / GLADIUS	AIRL without anchors	Reduced-form
1	0.000	0.000	0.000	0.000	0.000
2	0.000	0.000	0.000	0.025	0.082
3	0.000	0.000	0.000	0.017	0.094
5	0.000	0.000	0.000	0.015	0.123
7	0.000	0.000	0.000	0.016	0.129
10	0.000	0.000	0.000	0.014	0.139

Table 2: Type B counterfactual error relative to the oracle. All five non-NFXP methods are shown. Methods that recover the structural reward are exact at every k.

5.3 C. Shaping magnitude sweep

We construct shaped rewards $\tilde{r} = r^* + \alpha V^*(s) - \alpha\beta V^*(s')$ for $\alpha \in [0, 1]$ and evaluate the Type B counterfactual with $k = 3$. At $\alpha = 0$ the reward is exact and at $\alpha = 1$ it is maximally shaped.

α	Type B counterfactual error	Correlation with r^*
0.00	0.000	1.000
0.10	0.004	0.908
0.30	0.011	0.610
0.50	0.017	0.440
1.00	0.027	0.270

Table 3: Counterfactual error increases monotonically with shaping magnitude. Without anchors, the optimizer is not pinned down to a unique α.

5.4 D. Finite-sample convergence

We compare five finite-sample estimators across sample sizes from 200 to 10,000 trajectories (3 seeds each). The reduced-form estimator targets A^*, which is the wrong objective for Type B counterfactuals. IQ-Learn (χ^2) optimizes the Garg et al. (2021) tabular objective with L-BFGS-B. AIRL without anchors trains an adversarial discriminator without anchor constraints, so shaping ambiguity persists in finite samples as in Analysis A. AIRL with anchors enforces the two anchor constraints, with exit pinned to zero reward and the absorbing state s_{abs} pinned to zero value, achieving full identification at population. GLADIUS uses neural Q and EV networks with 150 training epochs.

N	Reduced-form	IQ-Learn (χ^2)	AIRL without anchors	AIRL with anchors	GLADIUS	NFXP
200	0.265	0.237	0.106	0.105	0.132	–
500	0.264	0.287	0.106	0.105	0.135	–
2,000	0.263	0.285	0.106	0.105	0.139	–
5,000	0.262	0.335	0.106	0.105	0.135	–
10,000	0.262	0.333	0.106	0.105	0.130	–

Table 4: Type B counterfactual error for $k = 3$ with 3 seeds. AIRL without anchors and AIRL with anchors achieve the lowest error, at 0.106 and 0.105. GLADIUS is close at about 0.13. Tabular IQ-Learn has local optima and degrades at large N. Reduced-form plateaus at 0.262 because it targets the wrong object. NFXP remains pending in this table.

5.5 E. Anchor misspecification

We set the true exit payoff to ε while the anchors assume it is zero. All five non-NFXP methods are evaluated at the population level. The reduced-form estimator and AIRL without anchors are not affected by the anchor misspecification because they do not use anchors. The error shown is for AIRL with anchors, which degrades smoothly.

ε	Correlation with r^*	Type B counterfactual error	In-sample π shift
0.0	1.000	0.000	0.000
0.5	0.976	0.002	0.000
1.0	0.909	0.004	0.000
2.0	0.729	0.008	0.000

Table 5: Anchor misspecification for AIRL with anchors. Degradation is smooth. In-sample CCPs are unaffected because the policy is identified regardless of reward levels.

5.6 F. Content-based paywall

High-content episodes become paid-only and the wait action is blocked, forcing buy-or-exit. This is a Type B shift that mirrors Section 7.3 of LSW. All five methods are evaluated at the population level.

Method	Counterfactual error
Oracle	0.000
AIRL with anchors	0.000
IQ-Learn / GLADIUS	0.000
AIRL without anchors	0.009
Reduced-form	0.185

Table 6: Content-based paywall counterfactual. The reduced-form estimator fails by 18.5 percentage points because it cannot separate content utility from continuation value.

5.7 G. State-only reward (Fu et al. transfer test)

Analyses A through F use action-dependent rewards, which is the LSW setting. Here we test the Fu et al. (2018) setting by replacing the true reward with a state-only function $r^*(s, a) = r^*(s)$ for all a, where $r^*(s) = u(c)$ depends only on content quality. This satisfies the conditions of Fu et al. Theorem 5.1, so AIRL with a state-only $g_\theta(s)$ should recover $r(s)$ up to a constant that cancels in the softmax. We evaluate the Type B counterfactual with $k = 3$, re-solving the Bellman equation without pinning $V(s_{\text{abs}}) = 0$ (the absorbing-state anchor is not part of the Fu et al. identification strategy).

Method	MSE vs. r^*	Correlation with r^*	Type B counterfactual error
Oracle / AIRL with anchors / IQ-Learn	0.000	1.000	0.000
AIRL with state-only reward ($g^* = r + \text{const}$)	2.778	1.000	0.000
AIRL without anchors ($\alpha = 0.5$)	13.689	0.530	0.026
Reduced-form	54.755	0.000	0.182

Table 7: State-only reward DGP. AIRL with state-only $g_\theta(s)$ recovers $r^*(s)$ up to a constant, so the MSE remains positive while the correlation is 1. It transfers perfectly and yields zero Type B counterfactual error. This is the mechanism behind the Fu et al. sim-to-real experiments. Generic AIRL without the state-only restriction still fails, with error 0.026, because the shaping residual is action-dependent. The reduced-form advantage has zero correlation with $r^*(s)$ because the true reward is identical across actions and the advantage reflects only continuation-value differences.

6 Summary

	Reduced-form	AIRL without anchors	AIRL with anchors	IQ-Learn	GLADIUS	NFXP
Uses transitions	No	No (simulated)	No (simulated)	Yes (known)	No (empirical next states)	Yes (known)
Recovers	A^*	r + shaping	r^*	r^*	r^*	r^*
Type A counterfactual	✓	✓	✓	✓	✓	✓
Type B counterfactual	×	×	✓	✓	✓	✓

Table 8: Summary of the six method variants considered in the appendix.

These results clarify the role of `econirl`. The package is not just a collection of estimators. It is a framework for deciding which estimators support which counterfactual claims. For in-sample fit and Type A state extrapolation, many methods can look equally successful because they recover the same within-state choice differences. For Type B counterfactuals that change transitions, delays,

or action sets, the package needs methods that recover a structural reward rather than a reduced-form advantage. That is why `econirl` supports several estimators under one interface and evaluates them not only by fit, but also by reward recovery and counterfactual validity.

References

Fu, J., Luo, K., and Levine, S. (2018). "Learning Robust Rewards with Adversarial Inverse Reinforcement Learning." ICLR.

Garg, D., Chakraborty, S., Cundy, C., Song, J., and Ermon, S. (2021). "IQ-Learn: Inverse soft-Q Learning for Imitation." NeurIPS.

Kalouptsidi, M., Scott, P. T., and Souza-Rodrigues, E. (2021). "Identification of Counterfactuals in Dynamic Discrete Choice Models." Quantitative Economics, 12(2), 351–403.

Kang, E. H., Yoganarasimhan, H., and Jain, L. (2025). "An Empirical Risk Minimization Approach for Offline Inverse RL and Dynamic Discrete Choice Models." arXiv:2502.14131.

Lee, P., Sudhir, K., and Wang, T. (2026). "Consumer Engagement with Sequential Content: A Content-Aware Dynamic Choice Model." Working paper, Yale University.

Magnac, T. and Thesmar, D. (2002). "Identifying Dynamic Discrete Decision Processes." Econometrica, 70(2), 801–816.

Ng, A. Y., Haley, D., and Russell, S. J. (1999). "Policy Invariance Under Reward Transformations." ICML.

Rust, J. (1987). "Optimal Replacement of GMC Bus Engines." Econometrica, 55(5), 999–1033.

Iskhakov, F., Jorgensen, R., Rust, J., and Schjerning, B. (2016). "Comment on Constrained Optimization Approaches to Estimation of Structural Models." Econometrica, 84(3), 997–1012.